Filippo Lollato

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Profile

Artificial Intelligence Systems MSc student and Computer Science graduate with academic project experience in computer vision, deep learning, and multi-agent software. Built and evaluated a multi-object tracking pipeline and developed incin, a Rust machine-learning framework exploring compile-time tensor-shape and invariant checks. Interested in taking ML systems from experimentation to reliable software products through integration, testing, and deployment.

Education

M.Sc. in Artificial Intelligence Systems *2025–26 to Present*
University of Trento *Trento, Italy*

- Current interests: integrating AI into software workflows and production services, agent systems, and AI infrastructure.

B.Sc. in Computer Science *2022–23 to 2024–25*
University of Trento *Trento, Italy*

- Thesis project: *incin* (originally titled *Kindle*), a Rust machine-learning framework exploring compile-time tensor shape and invariant checks.
- Relevant coursework: Advanced Algorithms and Data Structures, Systems Programming, Computer Architecture, Operating Systems, Linear Algebra, Probability and Statistics.

Experience

Teaching Assistant / Academic Tutor *2024–25 to Present*
University of Trento *Trento, Italy*

- Tutor undergraduate students in Rust, Advanced Programming, and Compilers. Started in the 2024/25 academic year, during the third year of my BSc.

Selected Projects

MOT Benchmark: Multi-Object Tracking *Python, PyTorch, MOT15/17/20, PPO*
Solo academic project comparing tracking-by-detection methods with a reinforcement-learning tracker.

- Implemented a shared evaluation pipeline for six trackers, including SORT, OC-SORT, ByteTrack, and MARL-MOT, across MOT15, MOT17, and MOT20.
- Compared results with CLEAR MOT and HOTA metrics; investigated evaluation and data issues to make benchmark results reproducible.

incin: Compile-Time Safety for Machine Learning Code *Rust, Machine Learning*
Rust machine-learning framework developed as my BSc thesis project.

- Explore compile-time checks for tensor shapes and invariants to make model code structurally safer and easier to reason about.
- Longer-term goal: help AI research agents produce valid ML code through clear architectural rules and compiler feedback.

ASA Autobots: Multi-Agent Delivery System *Node.js, BDI, PDDL, LLM Integration*
Team project combining agent coordination, language models, and symbolic planning.

- Co-led a two-person team building a delivery workflow with BDI agents, an LLM coordinator, and a PDDL planner.
- Worked on the system architecture and integration of agent components and service interfaces.

NLU: Intent Classification and Slot Filling *Python, PyTorch, BERT, Hugging Face, ATIS*
Natural language understanding project using a joint intent and slot-filling model.

- Built the training pipeline and evaluated the model on ATIS, achieving 0.9765 intent classification accuracy.

evolcpp *C++20, Templates, CMake*
In-progress C++ port of core abstractions from incin.

- Implement compile-time tensor shape checks using C++ templates.

Technical Skills

Computer Vision and Deep Learning: Multi-object tracking, tracking-by-detection, SORT, OC-SORT, ByteTrack, Kalman filtering, PyTorch, BERT, Hugging Face

ML Architectures and Methods: Attention layers, linear and sliding-window attention, modern transformer architectures, reinforcement learning (PPO)

AI Integration and Agent Systems: AI workflow integration, LLM orchestration, tool use and function calling, multi-agent systems, BDI, PDDL, streaming APIs (SSE, WebSockets), RAG

Backend, Infrastructure and Serving: Docker, REST APIs, Axum, Node.js, model serving (vLLM, Ollama, ONNX Runtime), model quantization (GGUF, AWQ, FP8, NVFP4)

Programming and Systems: Rust, C++20, Python, TypeScript, CUDA C/C++, Linux, CMake, Cargo, Git, CI/CD

Languages

English: Cambridge English B2 First, Grade A (CEFR C1)

Italian: Native